KAIZEN MATHS

Lesson starter

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Trapezium Rule

Name _____

S1

Use the trapezium rule with the ordinates given in the table.

Use the trapezium rule with h = 2 for the ordinates y = 11, 27, 51, 83, 123.

S2

Calculate the required ordinates, then apply the trapezium rule.

Use the trapezium rule with 4 strips to estimate $\int_0^4 (x^2 + 3)\,dx$.

S3

Find the step size or number of strips before applying the rule.

The interval [1, 8] is divided into 5 equal strips for the trapezium rule. Find the step size h and the number of ordinates used.

S4

Use the trapezium rule with the ordinates given in the table.

Use the trapezium rule with h = 1 for the ordinates y = 6, 11, 18, 27, 38.

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ALG-D1B0-L3 · S

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KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Trapezium Rule

Name _____

EA1

Calculate the required ordinates, then apply the trapezium rule.

Use the trapezium rule with 2 strips to estimate $\int_0^3 (x^2 + 1)\,dx$.

EA2

Find the step size or number of strips before applying the rule.

The interval [0, 7] is divided into 8 equal strips for the trapezium rule. Find the step size h and the number of ordinates used.

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KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Trapezium Rule

Name _____

EB1

Use the trapezium rule with the ordinates given in the table.

Use the trapezium rule with h = 1 for the ordinates y = 11, 18, 27, 38, 51.

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KAIZEN MATHS

Exit ticket

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Trapezium Rule

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Calculate the required ordinates, then apply the trapezium rule.

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ALG-D1B0-L3 · EA

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Trapezium Rule

Name _____

EB1

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Use the trapezium rule with h = 1 for the ordinates y = 11, 18, 27, 38, 51.

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Show your working

Teacher answers & working

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule · Trapezium Rule

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S1

Use the trapezium rule with the ordinates given in the table.

Use the trapezium rule with h = 2 for the ordinates y = 11, 27, 51, 83, 123.

Answer:

Estimate = 456.

Working:

For 4 strips, the trapezium rule is $\frac{h}{2}[y_0 + y_4 + 2(y_1 + y_2 + y_3)]$.

Substitute h = 2: $\frac{2}{2}[11 + 123 + 2(27 + 51 + 83)]$.

Interior sum: 27 + 51 + 83 = 161.

Estimate = $\frac{2}{2}[134 + 2(161)] = 456$.

S2

Calculate the required ordinates, then apply the trapezium rule.

Use the trapezium rule with 4 strips to estimate $\int_0^4 (x^2 + 3)\,dx$.

Answer:

Estimate = 34.

Working:

Step size h = $\frac{4-0}{4} = 1$.

The x-values are 0, 1, 2, 3, 4.

The y-values from $y = x^2 + 3$ are 3, 4, 7, 12, 19.

Use $\frac{h}{2}[y_0 + y_4 + 2(\text{sum of interior ordinates})]$.

Estimate = $\frac{1}{2}[3 + 19 + 2(23)] = 34$.

S3

Find the step size or number of strips before applying the rule.

The interval [1, 8] is divided into 5 equal strips for the trapezium rule. Find the step size h and the number of ordinates used.

Answer:

h = 1.4 and ordinates = 6.

Working:

Step size is interval width divided by number of strips.

$h = \frac{8-1}{5} = 1.4$.

If there are 5 strips, there is one more ordinate than the number of strips.

Number of ordinates = 5 + 1 = 6.

Teacher answers & working

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule · Trapezium Rule

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S4

Use the trapezium rule with the ordinates given in the table.

Use the trapezium rule with h = 1 for the ordinates y = 6, 11, 18, 27, 38.

Answer:

Estimate = 78.

Working:

For 4 strips, the trapezium rule is $\frac{h}{2}[y_0 + y_4 + 2(y_1 + y_2 + y_3)]$.

Substitute h = 1: $\frac{1}{2}[6 + 38 + 2(11 + 18 + 27)]$.

Interior sum: 11 + 18 + 27 = 56.

Estimate = $\frac{1}{2}[44 + 2(56)] = 78$.

EA1

Calculate the required ordinates, then apply the trapezium rule.

Use the trapezium rule with 2 strips to estimate $\int_0^3 (x^2 + 1)\,dx$.

Answer:

Estimate = 13.125.

Working:

Step size h = $\frac{3-0}{2}$ = 1.5.

The x-values are 0, 1.5, 3.

The y-values from $y = x^2 + 1$ are 1, 3.25, 10.

Use $\frac{h}{2}[y_0 + y_2 + 2(\text{sum of interior ordinates})]$.

Estimate = $\frac{1.5}{2}[1 + 10 + 2(3.25)] = 13.125$.

EA2

Find the step size or number of strips before applying the rule.

The interval [0, 7] is divided into 8 equal strips for the trapezium rule. Find the step size h and the number of ordinates used.

Answer:

h = 0.875 and ordinates = 9.

Working:

Step size is interval width divided by number of strips.

$h = \frac{7-0}{8} = 0.875$.

If there are 8 strips, there is one more ordinate than the number of strips.

Number of ordinates = 8 + 1 = 9.

Teacher answers & working

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EB1

Use the trapezium rule with the ordinates given in the table.

Use the trapezium rule with h = 1 for the ordinates y = 11, 18, 27, 38, 51.

Answer:

Estimate = 114.

Working:

For 4 strips, the trapezium rule is $\frac{h}{2}[y_0 + y_4 + 2(y_1 + y_2 + y_3)]$.

Substitute h = 1: $\frac{1}{2}[11 + 51 + 2(18 + 27 + 38)]$.

Interior sum: 18 + 27 + 38 = 83.

Estimate = $\frac{1}{2}[62 + 2(83)] = 114$.